

OPD Diagnostics

End-to-End Workflow & Test Plan

Slice 1 — OPD Pathology (Recommend Book Collect Result Verify Release)

Tenant	MBS Hospital
Module	Diagnostics (Pathology, OPD-sourced)
Scope	End-to-end workflow narrative + role-based UAT test cases
Prepared for	Internal UAT / Go-Live Readiness
Document version	1.0
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1. Purpose & Scope

This document describes the complete OPD (outpatient) pathology diagnostics workflow as actually implemented in SVK HMS, and provides a role-by-role set of test cases for User Acceptance Testing (UAT). It covers the full lifecycle of a lab test from the moment a doctor recommends it to the moment a verified report is released and its charge is posted to the patient's bill.

In scope: OPD-sourced pathology recommendations, booking & payment, sample collection (including batched multi-test collection), receipt at the lab, result entry, verification & release, and the supporting worklists/dashboards for each role.

Out of scope (see Section 7 for details): Imaging and Outsourced-lab workflows, IPD/Emergency-sourced recommendations, barcode label printing, and a handful of backend-only actions that have no frontend screen yet (decline/repeat recommendation, external report upload).

2. Test Accounts

ROLE	USERNAME	PASSWORD	USED FOR
Doctor	MBSDOC2026001	Test@12345	Recommending investigations
Reception	MBSRECP2026001	Test@12345	Booking, payment, catalogue linking
Phlebotomist	MBSPHLEB001	Test@12345	Sample collection
Lab Technician	MBSLABTECH001	Test@12345	Receiving samples, entering results
Pathologist	MBSPATH001	Test@12345	Verification & release
Admin	ADMIN-001	Admin@12345	Catalogue, rate lists, oversight (requires MFA)

All accounts are seeded, tenant-scoped test fixtures on the MBS Hospital tenant — not production credentials.

3. End-to-End Workflow Overview

This is the full journey of one test, from recommendation to a released report and posted bill charge:

1 Doctor recommends an investigation

DOCTOR

During or at the end of an OPD consultation, the doctor searches the catalogue (or adds a free-text test not yet catalogued) and saves it against the visit. This creates a **DiagnosticRecommendation** in **PENDING** status — a clinical statement, not yet a commitment or a bill.

2 Reception books & collects payment

RECEPTION

Reception sees the pending recommendation (via the Appointments queue "Tests to book" flag, the OPD visit's Investigations section, or the patient profile) and books it into a

DiagnosticOrder against a rate list, sets a collection date/window, and optionally takes payment. The recommendation flips to **BOOKED** .

3 Phlebotomist collects the sample

PHLEBOTOMIST

The test appears on the Collection Worklist. If payment is outstanding on an OPD order, collection is blocked (BR-04). The phlebotomist can select one or several due tests that share the same order, sample type and tube colour, and collect them together as a single physical sample — generating one accession number (e.g. **L-260811-0003**). Order items move to **COLLECTED** .

4 Lab technician receives the sample

LAB TECHNICIAN

The sample sits in "Awaiting Receipt" until a lab technician confirms its physical condition. Good condition moves it to **RECEIVED** ; a rejected sample (haemolysed/clotted/insufficient/contaminated) reverts to **ORDERED** and spawns a fresh collection task automatically — with no second charge to the patient (BR-05).

5 Lab technician enters results

LAB TECHNICIAN

Values are entered per parameter and instantly flagged (Normal/High/Low/Critical) against the matched reference range. A critical value immediately raises an alert to the ordering doctor (BR-08) — at entry time, not later. Once every parameter has a value, the item is submitted for verification (**RESULT_ENTERED**).

6 Pathologist verifies & releases

PATHOLOGIST

A different user from whoever entered the result (BR-03) reviews and verifies. This single action posts the bill charge (BR-01 — the only point money moves), generates and releases the PDF report, and marks the originating recommendation **COMPLETED** .

7 Report is live

ALL ROLES (view)

The report is now visible on the Reports screen (searchable by patient) and on the patient's profile, with full view/reprint/resend access logging.

4. Business Rules Reference

RULE	WHAT IT ENFORCES
BR-01	The bill charge for a test posts only when it is verified — never earlier.

BR-03	The user who entered a result cannot also verify it — a different pathologist login must do so.
BR-04	For OPD/Walk-in sourced orders, sample collection is blocked while the order is unpaid.
BR-05	A rejected sample gets a free recollection — no second charge is raised.
BR-08	A critical result value raises an alert automatically the instant it's entered.
BR-09	The price charged is frozen at order-booking time, even if the rate list changes later.
BR-10	A numeric test parameter needs at least one fallback reference range (Sex=Any, Age 0–40000 days) before the test can be activated.
BR-16	TPA/insurance-covered high-value items are gated on a recorded pre-authorization before collection.

5. Role-Based Test Cases

Each test case lists a precondition, the exact steps to perform, and the expected result. Use the blank Pass/Fail column to record UAT outcomes.

DOCTOR — recommending investigations

Login: MBSDOC2026001 / Test@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-DOC-01	Open a patient's OPD visit that is IN_CONSULTATION . In the Investigations section, search for a test that exists in the catalogue (e.g. "Complete Blood Count"), select it, set a priority, and complete the visit.	A new PENDING recommendation is created, linked to the catalogue test. It appears under "Already advised" on this visit and on the patient's profile (OPD tab) under "Investigations advised."	
TC-DOC-02	In the same Investigations search box, type a test name with no catalogue match and click "Add as free text."	Investigation is added tagged "Not in catalogue" (amber). After saving, the recommendation is created with freeTextName set and no linked test — it will show as unbookable on Reception's screen until an admin adds a matching catalogue test and it gets linked.	
TC-DOC-03	Add two or more investigations with different priorities (Routine/Urgent/STAT) on the same visit and save.	Each becomes its own recommendation tied to this visit, with its own priority preserved independently.	
TC-DOC-04	Open a new OPD visit for a patient who already has a pending recommendation from an earlier visit.	Expected (not a bug): The prior visit's recommendation does <i>not</i> appear under "Already advised" here — that list is scoped to the current visit only. It's still visible on the patient's profile across all visits.	

RECEPTION — booking, payment, catalogue linking

Login: MBSRECP2026001 / Test@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-REC-01	Precondition: a completed OPD visit has ≥1 pending investigation. Open Appointments Queue.	The patient's row shows a blue "N test(s) to book" badge and a "Book tests" button. The "Tests to book" filter chip shows the matching count.	

TC- REC-02	Click "Book tests." Select a rate list (e.g. Cash), set collection date/window/location, optionally enter a payment amount, click "Book."	A DiagnosticOrder is created (order no. DX-YYMMDD-NNNN), items in ORDERED status. The recommendation flips PENDING BOOKED . The test now appears on the Phlebotomist's Collection Worklist.
TC- REC-03	On the Book screen, attempt to select a free-text (not-in-catalogue) recommendation.	Checkbox is disabled with a "Not in catalogue" badge — cannot be booked as-is.
TC- REC-04	Precondition: admin has added a matching catalogue test for a previously free-text recommendation. Click "Link to catalogue" on that row, search, and select the match.	The recommendation's test link updates immediately; the row becomes checkable and bookable without a page reload.
TC- REC-05	Book an OPD-sourced order without collecting any payment. Do not pay it.	Order is created with paymentStatus = UNPAID . (See TC-PHL-04 for the resulting collection block.)

PHLEBOTOMIST — sample collection

Login: **MBSPHLEB001** / **Test@12345**

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC- PHL-01	Open Collection Worklist Due tab. Change the date picker to yesterday, then back to today.	Items due on or before the selected date are shown, oldest first. Anything still pending from a prior date carries forward into today's list with a red "Overdue" badge rather than disappearing.	
TC- PHL-02	Tick a single due test (paid order) and click "Collect."	A new accession number is generated (e.g. L-260811-0004); the item moves to COLLECTED ; it appears on the "Collected" tab tagged "Awaiting receipt" and on the Diagnostics dashboard's "Collected — Awaiting Receipt" tile.	
TC- PHL-03	Precondition: patient has 2 due blood tests from the same order needing the same tube colour. Tick both and click "Collect as one sample."	Both tests are linked to a single new accession number / sample record — one physical tube, one label to write. Both show together on the "Collected" tab.	
TC- PHL-04	Tick a due blood test, then try to also tick a due test needing a different tube colour or sample type (e.g. urine).	The second row's checkbox is disabled/greyed out — cannot be selected while an incompatible test is already selected.	
TC- PHL-05	Precondition: TC-REC-05 (unpaid OPD order). Attempt to collect it.	Blocked with "Payment is pending on this order — collection is blocked (BR-04)." No sample is created.	

TC- PHL-06	Open the "Collected" tab for today.	Every sample drawn today is listed with a status badge — Awaiting receipt (amber), Received (green), or Rejected (red) — so nothing "disappears" after collection.
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LAB TECHNICIAN — receipt & result entry

Login: MBSLABTECH001 / Test@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC- LAB-01	Open Result Entry. Review the "Awaiting receipt" list at the top.	Every collected-but-not-received sample is listed (patient, tests, accession no., time drawn) — browsable without needing to know the accession number.	
TC- LAB-02	Click a row in the "Awaiting receipt" list, select condition "Good," click "Confirm."	Sample moves to RECEIVED ; its order item(s) appear in the Result Entry worklist below, ready for value entry.	
TC- LAB-03	Receive a different sample with condition "Haemolysed" (or Clotted/Insufficient/ Contaminated).	Sample marked rejected (BR-05); order item reverts to ORDERED ; a fresh collection appointment is auto-created for the phlebotomist; no new charge is raised.	
TC- LAB-04	Expand a received test with configured parameters (e.g. CBC Hemoglobin). Enter a value within the normal reference range and Save.	Value saves; flag shows "Normal"; item status moves to IN_PROCESS on first save.	
TC- LAB-05	Enter a value at or beyond the configured critical threshold for a parameter.	Value is flagged Critical High/Low, and a CriticalAlert is automatically raised to the ordering doctor (BR-08) — immediately, not at verification.	
TC- LAB-06	With one or more parameters still blank, click "Submit for verification."	Blocked with an error listing exactly which parameter(s) are missing a value.	
TC- LAB-07	With every parameter filled in, click "Submit for verification."	Item status moves to RESULT_ENTERED ; it disappears from this worklist and appears on the Pathologist's Verification worklist.	
TC- LAB-08	Use the collapsed "Or look up by accession number" fallback with a known accession number and receive it from there instead of the list.	Same receive flow works via manual/typed or scanned accession entry.	

PATHOLOGIST — verification & release

Login: MBSPATH001 / Test@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
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TC-PATH-01	Open the Verification worklist.	Items with any abnormal result value are sorted to the top, ahead of fully-normal items.
TC-PATH-02	Open an item entered by the lab technician (TC-LAB-04-07) and click "Verify."	In one action: charge posts to the patient's bill (BR-01), a PDF report is generated and released (with QR verification code), item status moves VERIFIED RELEASED , and the originating recommendation flips to COMPLETED .
TC-PATH-03	Attempt to verify a result using the same login that entered it (if technically possible in your test setup, e.g. an Admin account that both entered and verifies).	Blocked with 403: "BR-03: the user who entered a result cannot verify it."
TC-PATH-04	Open an item in the Verification worklist and click "Reject to technician" with a reason.	Item reverts to IN_PROCESS and reappears on the Lab Technician's Result Entry worklist for correction.

ADMIN — catalogue & configuration

Login: **ADMIN-001** / **Admin@12345** (MFA required)

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-ADM-01	Diagnostics Catalogue "New Category." Create e.g. "Biochemistry."	New category available immediately in the Category dropdown when creating a test.	
TC-ADM-02	Click "New Test." Fill name, code, category, test type (Pathology), sample type (Blood), container colour, and save.	Test is created with isActive = false — it will show as "Inactive" and cannot yet be booked, per the toast: "add parameters, then activate it before it can be booked."	
TC-ADM-03	Open the new test "Add parameter." Set name, unit, result type = Numeric.	Parameter is added and listed under the test.	
TC-ADM-04	Attempt to activate the test before adding a reference range for its numeric parameter.	Blocked (BR-10): "Cannot activate — the following numeric parameters are missing a fallback reference range (sex=ANY, age 0-40000 days)."	
TC-ADM-05	Add a reference range with Sex = Any and Age 0–40000 days for the parameter, then activate the test again.	Activation succeeds; badge flips to "Active"; test becomes selectable for linking/ booking.	
TC-ADM-06	Diagnostics Rate Lists open "Cash" set a price for the newly activated test.	Price is saved; this is what gets frozen onto the order at booking time (BR-09).	

6. Cross-Cutting & Negative Test Cases

ID	SCENARIO	EXPECTED RESULT	PASS/FAIL
TC-NEG-01	Activate a test that has zero parameters configured (skip adding any).	Known edge case: activation succeeds trivially — BR-10's check only fires when numeric parameters exist. A test can go "Active" with nothing to actually enter a result against. Confirm this is acceptable, or treat as a follow-up fix.	
TC-NEG-02	As Reception, try to open <code>/diagnostics/collection</code> , <code>/diagnostics/results</code> , or <code>/diagnostics/verification</code> directly.	Frontend blocks with "access required" for that role, even though the backend API itself permits Reception read-only access to the collection worklist. Confirm this restriction is intentional.	
TC-NEG-03	As Admin, attempt to perform a clinical action directly — collect a sample, enter a result, or verify one.	Blocked by role, by design — Admin has configuration/oversight access, not clinical execution rights, mirroring real licensing separation of duties.	
TC-NEG-04	Attempt to batch-collect two order items belonging to two different orders for the same patient.	Rejected: "All order items in a single collection must belong to the same order."	
TC-NEG-05	Change a test's price on the rate list <i>after</i> an order was already booked at the old price, then verify that order's result.	The charge posted to the bill reflects the price frozen at booking time, not the updated rate list price (BR-09).	

7. Known Limitations / Out of Scope

- **Slice 2 (not yet built):** Imaging workflow, Outsourced-lab dispatch/return, full resource/machine scheduling, TPA/insurance pre-authorization activation across all test types.
- **Slice 3 (not yet built):** IPD/Emergency-sourced recommendations, critical-alert escalation, real SMS/email/WhatsApp notification service.
- **Barcode label printing:** not built. Accession numbers are generated digitally but must currently be hand-written onto physical containers. See separate discussion — architecture already supports adding this later without rework.
- **Backend-only, no frontend screen yet:** declining a recommendation ("going outside"), repeating a prior test, and uploading an external (outside-lab) report all have working API endpoints but no UI page calls them today.
- **Pending-recommendations follow-up worklist:** not built — nothing currently proactively surfaces recommendations that have sat `PENDING` for several days without being booked.
- **Partial payment top-up:** no UI exists yet to add further payment to an order that was booked with partial payment.